🛒 Shopper Spectrum: Customer Segmentation and Product Recommendation System:

1. Project Overview

Shopper Spectrum is an E-Commerce Analytics project designed to analyze customer purchasing behavior and generate personalized product recommendations. The project leverages RFM (Recency, Frequency, Monetary) Analysis, K-Means Clustering, and Item-Based Collaborative Filtering to identify customer segments and recommend relevant products.

The primary objective is to help businesses improve customer retention, optimize marketing strategies, and enhance customer experience through data-driven insights.

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2. Problem Statement

The e-commerce industry generates large volumes of transaction data daily. Understanding customer purchasing behavior is critical for improving customer engagement and increasing revenue.

This project aims to:

1. Analyze online retail transaction data.
2. Segment customers using RFM Analysis.
3. Identify High-Value, Regular, Occasional, and At-Risk customers.
4. Develop a product recommendation system using collaborative filtering techniques.
5. Deploy an interactive Streamlit application for real-time predictions and recommendations.

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3. Business Use Cases

- Customer Segmentation for Targeted Marketing Campaigns
- Personalized Product Recommendations
- Customer Retention and Loyalty Programs
- Inventory Planning and Demand Forecasting
- Data-Driven Decision Making

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4. Dataset Information

- **Dataset**

  Online Retail Dataset

- **Dataset Features**

Column	Description
InvoiceNo	Transaction Number
StockCode	Product Code
Description	Product Name
Quantity	Quantity Purchased
InvoiceDate	Date and Time of Purchase
UnitPrice	Product Price
CustomerID	Customer Identifier
Country	Customer Location

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5. Data Preprocessing

The following preprocessing steps were performed:

- Removed records with missing CustomerID values.
- Removed cancelled invoices.
- Removed transactions with zero or negative quantities.
- Removed transactions with zero or negative prices.
- Converted InvoiceDate into datetime format.
- Created TotalAmount feature.

- **Dataset Summary**

Metric	Count
Original Records	541,909
Cleaned Records	397,884
Removed Records	144,025
Unique Customers	4,338
Unique Products	3,877

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6. Exploratory Data Analysis (EDA)

The following analyses were performed:

- **Country Analysis**

Transaction volumes were analyzed across countries to identify major customer markets.

- **Top-Selling Products**

Top-performing products were identified based on purchase quantities.

- **Purchase Trend Analysis**

Monthly transaction trends were visualized to understand seasonal purchasing patterns.

- **Monetary Analysis**

Customer spending behavior was analyzed using transaction-level and customer-level monetary distributions.

- **RFM Distribution Analysis**

Recency, Frequency, and Monetary distributions were examined to understand customer purchasing behavior.

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7. Customer Segmentation

- ## RFM Feature Engineering

Three customer behavior metrics were calculated:

- **Recency**

Number of days since the customer's most recent purchase.

- **Frequency**

Number of transactions completed by a customer.

- **Monetary**

Total amount spent by a customer.

8. Data Transformation

- Log Transformation applied to reduce skewness.
- StandardScaler applied for normalization.

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9. Clustering Methodology

- **Algorithm Used**

  K-Means Clustering

- **Cluster Selection**

  The optimal number of clusters was determined using:

  - Elbow Method
  - Silhouette Score Analysis

- **Final Number of Clusters**

  4 Clusters

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10. Customer Segment Labels

Segment	Description
High-Value	Frequent, recent, and high-spending customers
Regular	Consistent and steady purchasers
Occasional	Customers with lower purchasing frequency
At-Risk	Customers inactive for a long period

- **Segment Distribution**

Segment	Customers
At-Risk	1,612
Regular	1,173
Occasional	837
High-Value	716

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11. Product Recommendation System

Recommendation Approach

Item-Based Collaborative Filtering

Methodology

1. Created Customer-Product Purchase Matrix.
2. Calculated Product Similarity using Cosine Similarity.
3. Recommended Top 5 similar products for the selected product.

Recommendation Output

The system returns five highly similar products based on customer purchase behavior.

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12. Model Evaluation

Clustering Evaluation

- Elbow Curve Analysis
- Silhouette Score Analysis

Recommendation Evaluation

- Cosine Similarity Matrix
- Product Similarity Heatmap

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13. Streamlit Application

The project includes an interactive Streamlit web application with two modules.

Customer Segmentation Module

Inputs:

- Recency
- Frequency
- Monetary

Output:

- High-Value
- Regular
- Occasional
- At-Risk

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14. Product Recommendation Module

Input:

- Product Name

Output:

- Top 5 Recommended Products

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15. Project Structure

```
shopper-spectrum-ecommerce
|
├── app.py
├── README.md
```

```
├── requirements.txt
│
├── data
│   └── online_retail.csv
│
├── models
│   ├── feature_columns.pkl
│   ├── kmeans.pkl
│   ├── scaler.pkl
│   ├── segment_mapping.pkl
│   └── product_list.pkl
│
├── notebook
│   └── Shopper_Spectrum.ipynb
│
│ ├── Project_Report
│   └── 🛒 Shopper Spectrum_Report.pdf
│
└── Screenshots
    ├── home_page1.png
    ├── home_page2.png
    ├── Customer_Segmentation.png
    ├── Customer_Segmentation_Graph.png
    ├── Product_Segmentation.png
    ├── Product_heatmap_similarity_matrix.png
    └── RFM_Distribution_Analysis.png
```

Note: product_similarity.pkl was excluded from GitHub because it exceeds GitHub's 100 MB file size limit. The recommendation model can be regenerated by running the notebook.

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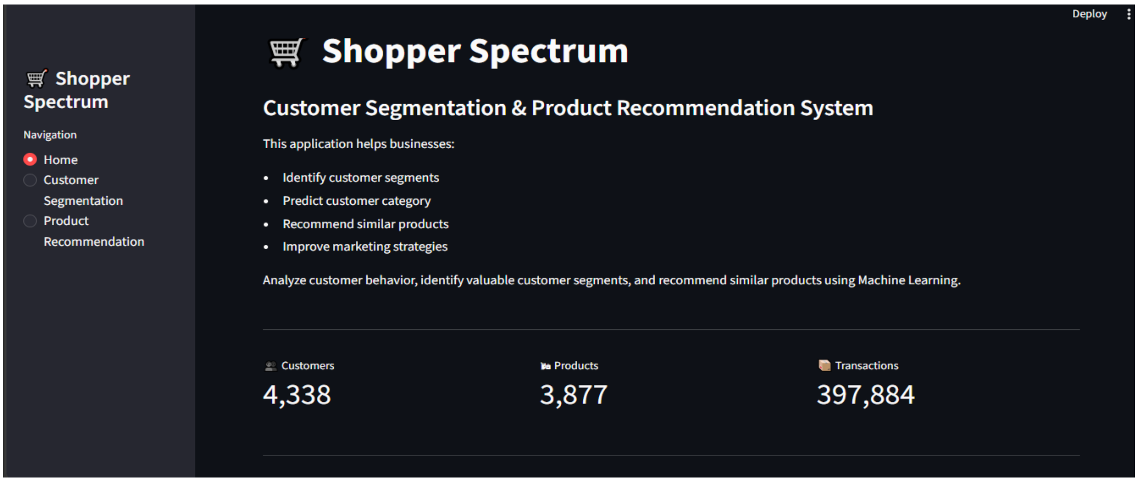
16. Project Screenshots

16.1 Home Page1

- **Home Page1**

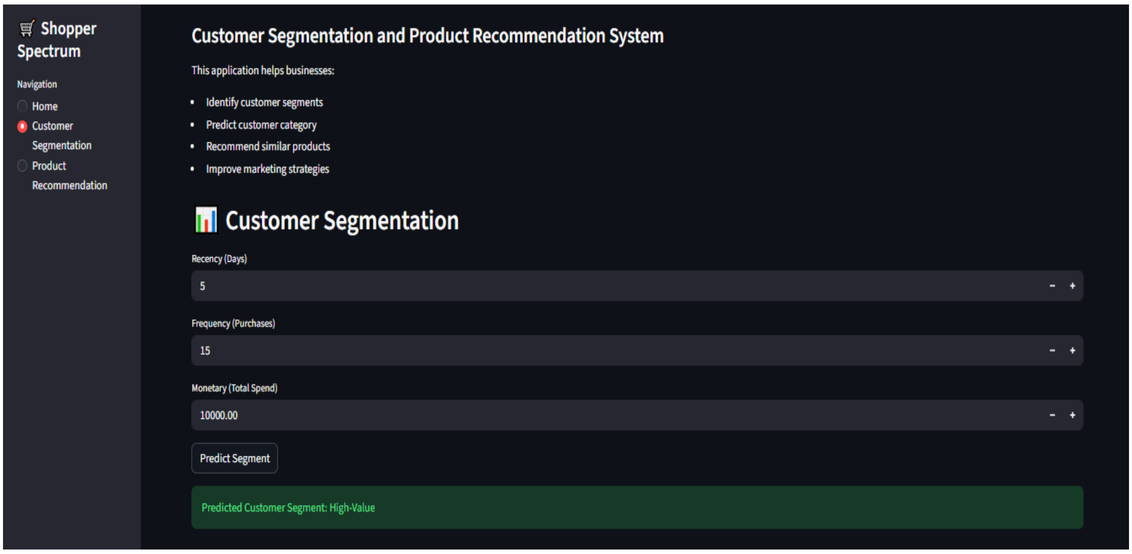


Description: The Home Page provides an overview of the Shopper Spectrum application. It displays project information, key business objectives, dataset statistics.

- **Home page2**

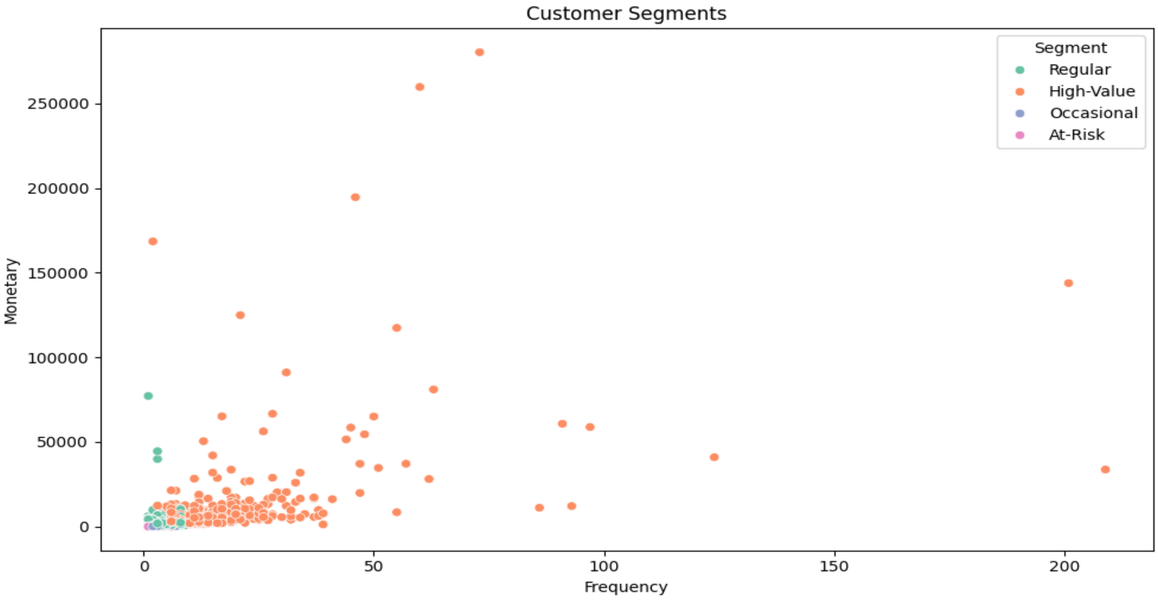
Description: customer segment distribution. This dashboard acts as the entry point for users to access customer segmentation and product recommendation modules.

16.2 Customer Segmentation



Description: This module allows users to enter customer behavior metrics including Recency, Frequency, and Monetary values. Based on the trained K-Means clustering model, the application predicts the customer's segment such as High-Value, Regular, Occasional, or At-Risk.

- **Customer Segmentation Graph**



Description: This visualization displays customer clusters generated using RFM Analysis and K-Means Clustering. It helps identify patterns among different customer groups and provides insights into customer purchasing behavior.

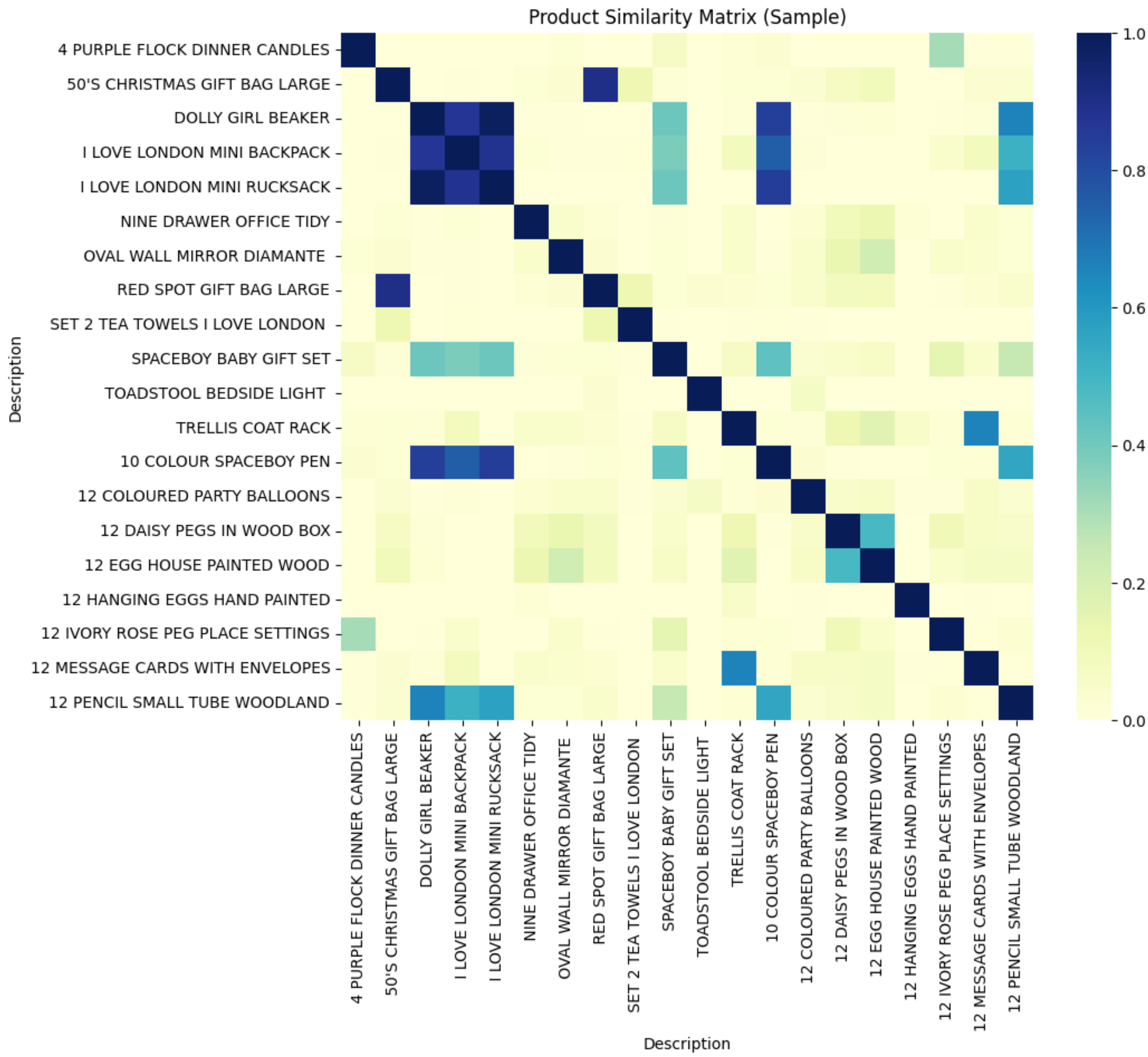
16.3 Product Recommendation

- **Product Segmentation**

Description: The Product Recommendation module enables users to select a product and receive the top five similar product recommendations. Recommendations are generated using Item-Based Collaborative Filtering and Cosine Similarity techniques.

16.4 Similarity Matrix

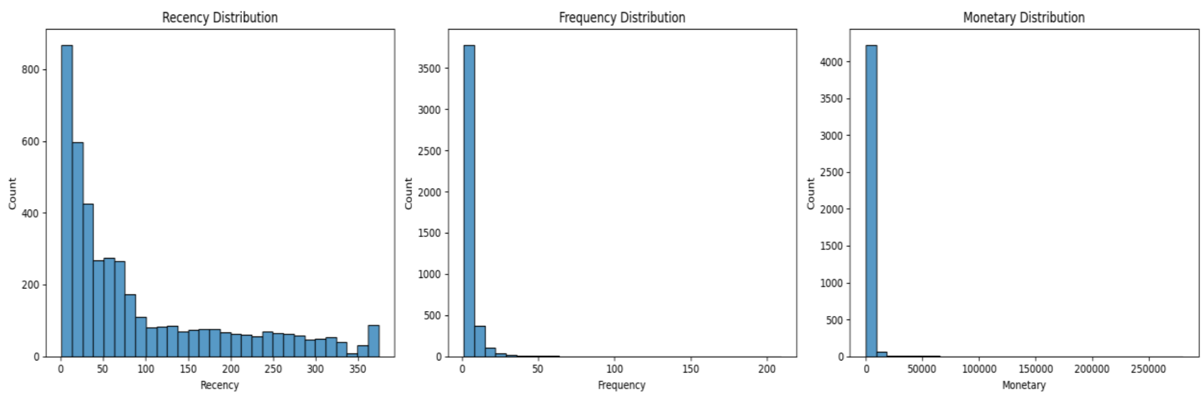
- **Product Heatmap Similarity Matrix**



Description: The heatmap visualizes product similarity scores calculated using cosine similarity. Products with higher similarity values indicate stronger relationships based on customer purchase history.

16.5 RFM Distribution Analysis

- **RFM Distribution Analysis**



Description: This visualization presents the distributions of Recency, Frequency, and Monetary values across customers. It helps understand customer behavior patterns and supports effective customer segmentation.

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17. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Streamlit
- Pickle

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18. Future Improvements

- Deploy application on Streamlit Cloud
- Use Hybrid Recommendation Systems
- Add Real-Time Customer Analytics Dashboard

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19. Conclusion

This project successfully implemented customer segmentation and product recommendation techniques using real-world e-commerce transaction data. RFM Analysis and K-Means Clustering enabled effective customer categorization, while Item-Based Collaborative Filtering generated meaningful product recommendations. The final Streamlit application provides an interactive interface for real-time customer segmentation and product recommendation, making it suitable for business intelligence and customer engagement use cases.

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